

LAB-4

Greedy: kruskals algorithm

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#include <stdio.h>

#define MAX 100

struct Edge {
    int u, v, weight;
};

int parent[MAX];

// Find the parent of a vertex
int find(int i) {
    while (parent[i] != i)
        i = parent[i];

    return i;
}

// Union two sets
void unionSet(int u, int v) {
    int rootU = find(u);
    int rootV = find(v);

    parent[rootV] = rootU;
}

// Sort edges in increasing order of weight
void sortEdges(struct Edge edges[], int e) {
    for (int i = 0; i < e - 1; i++) {
        for (int j = 0; j < e - i - 1; j++) {

            if (edges[j].weight > edges[j + 1].weight) {

                struct Edge temp = edges[j];
                edges[j] = edges[j + 1];
                edges[j + 1] = temp;
            }
        }
    }
}
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        edges[j + 1] = temp;
    }
}
}

int main() {

    int n, e;
    struct Edge edges[MAX];

    printf("Enter number of vertices: ");
    scanf("%d", &n);

    printf("Enter number of edges: ");
    scanf("%d", &e);

    printf("Enter edges (u v weight):\n");

    for (int i = 0; i < e; i++) {
        scanf("%d %d %d",
            &edges[i].u,
            &edges[i].v,
            &edges[i].weight);
    }

    // Initially, every vertex is its own parent
    for (int i = 0; i < n; i++) {
        parent[i] = i;
    }

    // Sort edges by weight
    sortEdges(edges, e);

    printf("\nEdges selected for MST:\n");

    int count = 0;
    int totalWeight = 0;

    // Kruskal's algorithm
    for (int i = 0; i < e && count < n - 1; i++) {

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    int u = edges[i].u;
    int v = edges[i].v;

    // If they belong to different sets,
    // adding this edge will not create a cycle
    if (find(u) != find(v)) {

        printf("%d -- %d = %d\n",
            u, v, edges[i].weight);

        totalWeight += edges[i].weight;

        unionSet(u, v);

        count++;
    }
}

// Check if MST was successfully formed
if (count != n - 1) {

    printf("\nMST cannot be formed.\n");
    printf("The graph is disconnected.\n");

}
else {

    printf("\nTotal weight of MST = %d\n", totalWeight);
}

return 0;
}

```

OUTPUT-

```
Enter number of vertices: 5
Enter number of edges: 8
Enter edges (u v weight):
0 1 2
0 2 1
1 4 3
4 3 4
3 2 8
2 1 9
0 3 10
1 3 11

Edges selected for MST:
0 -- 2 = 1
0 -- 1 = 2
1 -- 4 = 3
4 -- 3 = 4

Total weight of MST = 10
```